

S.T.A.R. Experiment Registry

Version 0.1 — Priority A Experimental Baseline

Controlled by S.T.A.R. Research Charter v0.2

Derived from Claim/Evidence Registry v0.1

1. Purpose

The S.T.A.R. Experiment Registry converts claims in the Claim/Evidence Registry into executable, falsifiable experiments.

The Registry answers:

Exactly what experiment will test each claim, using what data, what mathematical construction, what controls, what null model, and what constitutes success or failure?

The Registry is deliberately stricter than the historical ACSC experimental descriptions.

Historical experiments remain preserved, but a new experiment must distinguish:

[
 \text{historical demonstration}
]

from

[
 \text{preregistered validation experiment}.
]

2. Governing Experimental Principle

Every experiment must follow:

```
[
\boxed{
\text{Claim}
\rightarrow
\text{Prediction}
\rightarrow
\text{Dataset}
\rightarrow
\text{Locked Construction}
\rightarrow
\text{Null}
\rightarrow
\text{Test}
\rightarrow
\text{Independent Evaluation}
}
]
```

The experiment may not be modified after inspection of the primary test results unless the modification is recorded as a new exploratory experiment.

3. Experiment Status Vocabulary

Status	Meaning
PROPOSED	Experiment specified but not yet locked
PREREGISTERED	Protocol locked before execution
ACTIVE	Execution underway
COMPLETED	Execution finished
REPLICATING	Independent replication underway
SUPPORTED	Primary endpoint passed preregistered criteria
NOT SUPPORTED	Primary endpoint failed
INCONCLUSIVE	Data/protocol insufficient

FALSIFIED	Specific claim contradicted
EXPLORATORY	Result generated outside preregistered protocol

4. Standard Experiment Record

Every experiment shall contain:

Field	Required
Experiment ID	Yes
Claim ID	Yes
Experiment Name	Yes
Scientific Question	Yes
Primary Hypothesis	Yes
Alternative Hypothesis	Yes
Dataset	Yes
Population	Yes
Inclusion Criteria	Yes
Exclusion Criteria	Yes
Arithmetic Dataset	If applicable
Astronomical Dataset	If applicable
Mapping	Yes
Mapping Version	Yes
Locked Parameters	Yes
Derived Parameters	Yes
Null Model	Yes

Alternative Controls	Yes
Data Split	Yes
Primary Endpoint	Yes
Secondary Endpoints	Preferred
Statistical Test	Yes
Effect Size	Yes
Significance Criterion	Yes
Multiple-Test Correction	If applicable
Success Condition	Yes
Failure Condition	Yes
Falsification Condition	Yes
Reproducibility Requirement	Yes
Code Version	Required at execution
Random Seed	Required where applicable
Computational Environment	Required
Blind/Unblind Status	Yes
Result	After execution
Interpretation	After execution
Registry Status	After execution

5. Global Experimental Standards

5.1 No post-hoc threshold selection

A threshold discovered after seeing the result cannot be presented as the primary criterion.

5.2 Historical thresholds are controls, not acceptance standards

The historical:

[
 $W_2 < 0.01$
]

criterion is retained because it is part of the ACSC record.

It is **not** automatically accepted as the scientific decision threshold.

The primary modern evaluation shall use a null distribution and effect-size comparison.

5.3 Alternative mappings are mandatory

Whenever (Φ) is tested, at least one non- (Φ) mapping must be tested under comparable conditions.

The principal alternative family is:

- MCJ;
- PTD;
- FT.

These are explicitly identified in the alternative-mapping document as BSD-independent alternatives.

5.4 Rank cannot be evaluated in isolation

For rank-based experiments, the minimum arithmetic comparison set is:

[
 r ,
 $\log\{\text{Reg}\}$,
 $\log N$,
]

$$\frac{\log|\Delta|}{T}$$

Where data availability permits, the composite feature vector is:

$$[X_E, (r_E, \log\{\text{Reg}\}_E, \log N_E, \log|\Delta_E|, T_E)]$$

5.5 Significance is not predictive utility

Every predictive experiment records both:

$$[\text{statistical significance}]$$

and

$$[\text{out-of-sample predictive utility}]$$

A statistically significant but practically negligible effect is recorded as such.

6. Priority A Experiment Map

Experiment	Primary Claim
EXP-A01	CROSS-001 — Rank ↔ cosmic-web morphology
EXP-A02	CROSS-002 — Rank vs. other arithmetic invariants

EXP-A03	ACSC-002 — Arithmetic $\leftrightarrow$ cosmic topology
EXP-A04	MAP-006 — Alternative mapping controls
EXP-A05	CROSS-005 — Arithmetic structure $\leftrightarrow$ SFR
EXP-A06	ACSC-008 — Rank $\leftrightarrow$ proposed structural dimensionality
EXP-A07	ACSC-002/005 — Scaling/convergence robustness

The experiments are deliberately separated so that a positive result in one does not automatically count as evidence for every other claim.

7. EXP-A01 — Rank–Cosmic-Web Morphology

Experiment ID

[
 $\boxed{\text{EXP-A01}}$
]

Claim

CROSS-001

Arithmetic rank may be related to cosmic-web morphology.

Scientific Question

Does elliptic-curve rank contain statistically reproducible information about whether an astronomical environment is classified as cluster, filament, or void?

Primary Hypothesis

[
H_1:
 $P(r|\text{mid } C, F, V)$
]

differs systematically from the appropriate null distribution.

Null Hypothesis

[
H_0:
 r
 $\perp$
 $\text{cosmic-web environment}$
]

after controlling for relevant observational and sampling covariates.

Astronomical Dataset

Primary candidate:

SDSS galaxy catalog / volume-limited sample.

The ACSC source specifically proposes SDSS galaxy coordinates and a volume-limited sample.

Dataset version

TBD – exact SDSS release/catalog must be frozen before preregistration.

Population

A predefined volume-limited galaxy sample.

Required fields

At minimum:

- RA;

- Dec;
- redshift;
- galaxy identifier.

Additional fields should be retained for covariate control.

Cosmic-Web Classification

The classification pipeline must be frozen before rank analysis.

Cluster

Use an independently defined cluster/group catalog or cluster-density classification.

Filament

Use primarily:

- geometric skeleton;
- graph;
- filament-finder methodology.

Persistent homology may provide supporting topological characterization but shall not itself define “filament = (β_1).”

Void

Use an independently defined void catalog or void-finding algorithm.

Arithmetic Dataset

Candidate source:

- LMFDB;
- Cremona database;
- independently generated elliptic curves where required.

The ACSC monograph specifically identifies LMFDB and Cremona as arithmetic sources.

Arithmetic Variables

Primary:

```
[  
r_E.  
]
```

Secondary:

```
[  
\log\operatorname{Reg}(E),  
\quad  
\log N_E,  
\quad  
\log|\Delta_E|,  
\quad  
T_E.  
]
```

Mapping

For the rank-specific experiment:

```
[  
z_E=200r_E  
]
```

may be retained as the historical mapping, but it is not itself treated as physical.

Critical Matching Problem

The experiment must define how astronomical objects are assigned arithmetic objects.

This cannot be chosen after observing the results.

Matching protocol

TBD – must be preregistered.

Possible choices include:

1. rank distribution comparison without one-to-one object matching;
2. fixed stochastic assignment;
3. deterministic binning;
4. projection-space density comparison.

A direct galaxy-to-curve identity should **not** be assumed.

Null Model

Primary:

[
H_0^{\rm perm}
]

randomly permute rank labels across astronomical environments while preserving the observed rank distribution.

Secondary nulls:

- spatial permutation;
 - redshift-stratified permutation;
 - density-matched permutation.
-

Locked Parameters

Before execution:

- sample definition;
- redshift limits;
- environment classifier;
- rank computation/version;
- arithmetic sample;
- matching rule;
- spatial smoothing scale;
- binning;
- random seed;
- statistical test.

Train/Test Design

This experiment is primarily inferential.

Use:

- discovery/training partition for any model selection;
- held-out test partition for final association.

No environment classifier may be tuned using rank.

Primary Endpoint

Effect of rank on environment classification after covariate adjustment.

Candidate primary statistic:

$\text{performance}_{\text{rank}}$

$\text{performance}_{\text{null}}$.
]

Exact metric:

TBD – preregister before execution.

Success Condition

A successful result requires:

1. statistically significant deviation from the primary null;
 2. non-negligible effect size;
 3. persistence across reasonable observational controls;
 4. replication on held-out data;
 5. no equivalent or stronger explanation from obvious covariates.
-

Failure Condition

Failure occurs if:

- rank does not outperform null;
 - the effect disappears after controls;
 - effect is confined to training data;
 - effect is attributable to survey geometry or selection;
 - alternative randomization reproduces the effect.
-

Reproducibility

Must include:

- raw-data identifiers;
 - preprocessing script;
 - exact rank source/version;
 - random seed;
 - classifier version;
 - environment-definition file;
 - immutable result table.
-

8. EXP-A02 — Rank Versus Other Arithmetic Invariants

Experiment ID

[
\boxed{\text{EXP-A02}}
]

Claim

CROSS-002

Rank is more informative than other arithmetic invariants.

This is a comparative hypothesis, not an assumption.

Primary Question

Does rank provide information about cosmic structure that cannot be explained equally well by:

[
 $\log N$,
 $\log |\Delta|$,
 $\log \operatorname{Reg}$,
 T ?
]

Null

[
 H_0 :
 $\text{rank provides no incremental predictive information}$.
]

Models

Model A

[
 $Y=f(r)$.
]

Model B

[
 $Y=f(\log N, \log |\Delta|, \log \operatorname{Reg}, T)$.
]

Model C

[
Y=f(r,\log N,\log|\Delta|,
\log\operatorname{Reg},T).
]

Primary Test

Compare out-of-sample predictive performance.

The critical quantity is:

$\text{Score}(C)$

$\text{Score}(B)$.
]

Null Model

Permutation of rank conditional on the other arithmetic invariants.

Train/Test

Recommended:

[
60%/20%/20%
]

train/validation/test.

However, the exact split must be frozen before execution.

For grouped astronomical data, splits must prevent spatial leakage.

Success

Rank demonstrates reproducible incremental predictive information on the untouched test set.

Statistical significance alone is insufficient.

Failure

Rank adds no meaningful out-of-sample information.

9. EXP-A03 — Arithmetic/Cosmic Persistent-Homology Correspondence

Experiment ID

[
\boxed{\text{EXP-A03}}
]

Claim

ACSC-002

Arithmetic projection produces a topological signature that corresponds to cosmic structure.

Historical Construction

\left(
\log_{10}|\Delta|,
\log_{10}N,
200r

\right).
]

This is explicitly the projection used in the ACSC monograph.

Astronomical Dataset

Primary:

SDSS volume-limited 3D galaxy sample.

Secondary:

- cosmological simulation;
- independent survey.

The ACSC source proposes SDSS, 2dF, Euclid, and Millennium Simulation as possible observational/simulation sources.

Arithmetic Dataset

Primary:

LMFDB/Cremona elliptic curves with required invariants.

Primary Topological Objects

[
 $D_k^{\rm arith}$,
 $\sqcup$
 $D_k^{\rm cosmic}$,
 $\sqcup$
 $k=0,1,2$.
]

Primary Comparison

Wasserstein distance:

[
W_p(D_k^{\rm arith}, D_k^{\rm cosmic}).
]

Historical Criterion

The source specifies:

[
W_2 < 0.01
]

as a historical threshold.

This is retained as:

[
\boxed{\text{Historical Criterion H-W2}}
]

but is **not** the primary modern acceptance threshold.

Primary Null

Generate randomized arithmetic projections preserving:

- number of points;
- coordinate marginal distributions;
- relevant scaling;
- sample size.

The cosmic dataset should similarly be subjected to spatial nulls where appropriate.

Alternative Nulls

N1 — Coordinate permutation

Shuffle arithmetic invariant associations.

N2 — Rank permutation

Preserve $(\log N)$ and $(\log|\Delta|)$, randomize rank.

N3 — Random projection

Generate an equivalent-dimensional random mapping.

N4 — Alternative arithmetic mapping

Use MCJ/PTD/FT.

Critical Normalization Rule

Wasserstein distances are sensitive to scale.

Therefore:

Arithmetic and astronomical point clouds must be normalized using a procedure frozen before comparison.

No normalization may be selected because it minimizes (W_2) .

Primary Endpoint

The primary endpoint is not simply:

[
 $W_2 < 0.01$.
]

It is:

$$\frac{W_{\text{null}} - W_{\text{observed}}}{\{$$

$$\sigma_{\text{null}}$$

or an equivalent preregistered null-relative statistic.

Success

Observed correspondence must outperform the null distribution with:

- statistically significant separation;
 - meaningful effect size;
 - robustness to normalization;
 - persistence across reasonable sample sizes;
 - independent replication.
-

Failure

ACSC correspondence is not supported if the observed topology is statistically indistinguishable from null mappings or if the correspondence disappears under reasonable normalization choices.

Important Filament Rule

$$\boxed{\beta_1 \neq \text{“number of filaments” by definition}}.$$

Filament correspondence must additionally be evaluated through geometric/graph/skeleton structure.

10. EXP-A04 — Alternative Mapping Control Experiment

Experiment ID

[
\boxed{\text{EXP-A04}}
]

Claim

MAP-006

MCJ, PTD, and FT provide meaningful BSD-independent controls for (Φ) .

Purpose

Determine whether apparent success is specific to the proposed arithmetic structure or is produced generically by low-dimensional projections.

Mapping Families

Primary

[
 Φ :
 $(\log|\Delta|,$
 $\log N,$
 $\alpha_r)$.
]

MCJ

(j) -invariant-based mapping.

PTD

Purely arithmetic mapping.

FT

Frobenius-trace-based mapping.

The alternative-mapping document explicitly proposes these families as alternatives to the primary (Φ) construction.

Locked Principle

All mappings must receive:

- identical sample size;
 - identical observational target;
 - identical topology algorithm;
 - identical distance metric;
 - identical normalization procedure;
 - identical null-generation procedure.
-

Null

Random mapping preserving coordinate distributions.

Primary Endpoint

Relative null-adjusted topological correspondence:

$\text{performance}(m)$

$E[\text{performance}(\text{null})]$
]

Success

The result supports mapping specificity only if:

[
S_\\Phi
]

is reproducibly distinguishable from both null mappings and alternative arithmetic mappings.

Failure

If all mappings perform similarly, the evidence supports a generic projection/topology effect rather than specificity of $(\\Phi)$.

This is not necessarily a failure of topology itself; it is a failure of the claim that the particular arithmetic construction is responsible.

11. EXP-A05 — Arithmetic Structure and Star Formation Rate

Experiment ID

[
\\boxed{\\text{EXP-A05}}
]

Claim

CROSS-005

Arithmetic/topological structure may contain information about star formation rate.

Scientific Role

This experiment is the primary astrophysical laboratory for S.T.A.R.

It is deliberately treated as an **independent empirical test**, not as proof of ACSC.

Candidate Astronomical Dataset

The existing S.T.A.R. data pipeline includes:

- SDSS;
- MaNGA HI;
- Pipe3D;
- GEMA;
- MagPhys;
- StellarMasses;
- EnvironmentMeasures;
- Galaxy classifications;
- Galaxy Zoo;
- photometric data.

The exact final dataset must be frozen before execution.

Target

Primary:

[
 $Y = \log\{\rm SFR\}$
]

where a directly measured SFR is available.

If a proxy is used:

[
 $Y_{\rm proxy}$
]

must be explicitly labeled as such.

A proxy may not be presented as a measured SFR.

Candidate Features

Arithmetic:

[
r,\quad
\log N,\quad
\log|\Delta|,
\quad
\log\operatorname{Reg},
\quad
T.
]

Astronomical controls may include:

- stellar mass;
- gas mass;
- metallicity;
- redshift;
- morphology;
- environment;
- photometric variables.

Critical Data Rule

No SFR-derived quantity may appear among the input features.

No variable that directly encodes the target may leak into the training set.

Models

Baseline

Standard astrophysical predictors.

Arithmetic-only

```
[  
Y=f(X_{\rm arith}).  
]
```

Combined

```
[  
Y=f(X_{\rm astro},X_{\rm arith}).  
]
```

Primary Test

Does adding arithmetic features improve out-of-sample prediction beyond the astrophysical baseline?

$R^2_{\rm combined}$

```
R^2_{\rm baseline}.  
]
```

Other metrics:

- MAE;
- RMSE;
- calibration;
- rank correlation.

Exact primary metric must be locked before execution.

Null Model

Permute arithmetic features across galaxies while preserving their marginal distributions.

Secondary:

- redshift-stratified permutation;
 - mass-stratified permutation;
 - spatial permutation.
-

Train/Test Design

Spatial/grouped split preferred.

A galaxy from the same structural group should not appear simultaneously in training and test when that could create leakage.

Recommended starting design:

```
[  
60/20/20  
]
```

train/validation/test.

Exact split frozen before model training.

Model Discovery

The existing S.T.A.R. methodology may use:

- Random Forest;
- Gradient Boosting;
- symbolic regression;
- SHAP.

But:

```
[  
\boxed{  
\text{ML discovery}\neq\text{physical proof}.  
}  
]
```

Symbolic regression outputs become candidate equations requiring independent validation.

Success

Arithmetic features produce a reproducible improvement in held-out prediction after:

- astrophysical controls;

- null comparison;
 - leakage testing;
 - independent validation.
-

Failure

No improvement over the baseline or improvement disappears under null/control tests.

Stronger Success

An arithmetic feature set remains useful across:

- different galaxy populations;
 - independent survey subsets;
 - independent model families.
-

12. EXP-A06 — Rank and Structural Dimensionality

Experiment ID

```
[  
\boxed{\text{EXP-A06}}  
]
```

Claim

ACSC-008

Rank may correspond to different classes of cosmic structure.

Historical Hypothesis

[
 $r=0 \rightarrow \text{void-like},$
]

[
 $r=1 \rightarrow \text{filament-like},$
]

[
 $r=2 \rightarrow \text{cluster/surface-like},$
]

[
 $r=3 \rightarrow \text{higher-order structure}.$
]

These are treated as hypotheses, not definitions.

Primary Test

Estimate:

[
 $P(C \mid r),$
 $\quad$
 $P(F \mid r),$
 $\quad$
 $P(V \mid r).$
]

Null

Rank labels randomly assigned conditional on the observed rank distribution.

Controls

At minimum:

- redshift;
 - galaxy density;
 - stellar mass;
 - survey selection.
-

Primary Endpoint

Pre-registered association statistic between rank and independently classified environment.

Possible choices:

- mutual information;
- multinomial likelihood ratio;
- permutation statistic.

Exact primary statistic must be frozen before analysis.

Success

A reproducible rank/environment relationship survives:

1. permutation null;
 2. observational controls;
 3. independent sample.
-

Failure

No association beyond null expectation.

Strong Falsification

If rank-specific predictions consistently point in the opposite direction from the proposed mapping, the specific rank-dimensionality formulation should be marked:

[
\\boxed{\\text{FALSIFIED}}
]

even if some other arithmetic/cosmic relationship survives.

13. EXP-A07 — Scaling and Convergence

Experiment ID

[
\\boxed{\\text{EXP-A07}}
]

Claims

ACSC-002

and

ACSC-005

Question

Does arithmetic/cosmic correspondence become more stable as the arithmetic sample grows?

Historical Prediction

[
W_2
\\left(
D_k^{\\rm arith}(n),
D_k^{\\rm cosmic}
\\right)
\\rightarrow

$\text{\text{smaller values}}$
]

as:

[
 $n \rightarrow \infty$.
]

Experimental Design

Construct arithmetic samples of increasing size:

[
 $n_1 < n_2 < n_3 < \dots < n_m$.
]

The sample-selection algorithm must be fixed.

Critical Control

Sample density itself can produce apparent topological convergence.

Therefore compare against:

- random point clouds;
 - density-matched random clouds;
 - alternative arithmetic mappings.
-

Primary Endpoint

Slope of the learning/convergence curve:

[
 $\frac{dW_2}{d \log n}$.
]

Success

A statistically reproducible improvement with increasing (n) survives density-matched nulls.

Failure

Improvement disappears when point density is controlled.

14. Experiment-Level Multiple-Hypothesis Control

The Priority A experiments contain related hypotheses.

The Registry therefore distinguishes:

Primary confirmatory endpoints

One per experiment.

Secondary endpoints

Additional measurements.

Exploratory endpoints

Anything discovered after seeing the data.

Exploratory findings may be scientifically valuable but cannot be promoted to confirmatory findings retroactively.

15. Parameter Lock Registry

Every experiment must have a corresponding parameter-lock record.

Minimum fields:

Parameter	Required
Dataset version	Yes
Arithmetic database version	Yes
Curve inclusion range	Yes
Conductor limit	Yes
Rank computation method	Yes
Mapping formula	Yes
Scaling constants	Yes
Coordinate normalization	Yes
TDA algorithm	Yes
Filtration type	Yes
Homology dimensions	Yes
Distance metric	Yes
Null-generation method	Yes
Random seed	Yes
Train/test split	Yes
Model hyperparameters	Yes
Statistical test	Yes
Primary threshold	Yes

16. Blind Analysis Standard

Where feasible:

Stage 1

Freeze:

- data;
- preprocessing;
- mappings;
- nulls;
- primary endpoint.

Stage 2

Generate analysis outputs without interpreting them.

Stage 3

Unblind the primary endpoint.

Stage 4

Run secondary analyses.

Stage 5

Run exploratory analyses.

17. Reproducibility Standard

A completed experiment must be reproducible from:

```
[  
  \boxed{  
    \text{Data Identifier}  
    +  
    \text{Code Version}  
    +  
    \text{Parameters}  
    +  
    \text{Environment}  
    +  
    \text{Seed}  
  }  
]
```

The minimum reproducibility package contains:

1. dataset identifiers;
 2. preprocessing code;
 3. arithmetic-generation code;
 4. mapping implementation;
 5. TDA implementation;
 6. statistical analysis;
 7. configuration file;
 8. random seeds;
 9. software versions;
 10. output tables;
 11. experiment log.
-

18. Experiment Result Classification

A completed experiment receives one of the following outcomes.

PASS — Statistical Correspondence

The predicted relationship is statistically distinguishable from the null.

This does **not** imply physical causation.

PASS — Predictive Utility

The relationship improves independent out-of-sample prediction by a practically meaningful amount.

FAIL — Null Consistent

The observed relationship is consistent with the null.

FAIL — Alternative Explanation

The relationship disappears after controls or is equally explained by an alternative mapping.

FAIL — Specific Prediction

The particular prediction is contradicted.

INCONCLUSIVE

Data quality, computational limitations, or insufficient sample size prevent a valid conclusion.

19. Important Separation of Outcomes

The following statements are not equivalent:

[
 $\text{Rank correlates with structure}$
]

[
 $\neq$
]

[
 $\text{Rank predicts structure}$
]

[
 $\neq$
]

[
 $\text{Rank causes structure}$
]

[
\neq
]

[
\text{Rank is gravity}.
]

Likewise:

[
\text{Arithmetic topology resembles cosmic topology}
]

does not establish:

[
\text{the universe is generated by arithmetic}.
]

20. Priority A Execution Order

The experiments should proceed in this order:

Phase A — Controls

[
\boxed{\text{EXP-A04}}
]

Alternative mappings.

Phase B — Rank

[
\boxed{\text{EXP-A01}}
]

[
\boxed{\text{EXP-A06}}
]

Rank/environment relationships.

Phase C — Comparative invariants

[
\\boxed{\\text{EXP-A02}}
]

Rank versus other arithmetic variables.

Phase D — Topological correspondence

[
\\boxed{\\text{EXP-A03}}
]

Arithmetic versus cosmic persistence structure.

Phase E — Convergence

[
\\boxed{\\text{EXP-A07}}
]

Sample-size/scaling behavior.

Phase F — Astrophysical laboratory

[
\\boxed{\\text{EXP-A05}}
]

Arithmetic/SFR prediction.

This ordering prevents the topological experiment from becoming the sole test of the theory and provides independent astrophysical evidence streams.

21. Minimum Evidence Required Before Physical Interpretation

No Priority A result should be promoted into the Physical Correspondence Registry unless it satisfies at least:

```
[
\boxed{
\text{S3 — reproducible statistical correspondence}
}
]
```

and preferably:

```
[
\boxed{
\text{S4 — independent replication}
}
]
```

For claims involving predictive astrophysics:

```
[
\boxed{
\text{S5 — demonstrated out-of-sample predictive utility}
}
]
```

is required before strong predictive language is used.

22. Current Experiment Status

ID	Claim	Current Status	Primary Role
EXP-A01	Rank $\leftrightarrow$ cosmic-web morphology	PROPOSED	Primary empirical test
EXP-A02	Rank vs. other invariants	PROPOSED	Feature specificity
EXP-A03	Arithmetic $\leftrightarrow$ cosmic topology	PROPOSED	Core ACSC test
EXP-A04	Alternative mappings	PROPOSED	Critical control
EXP-A05	Arithmetic $\leftrightarrow$ SFR	PROPOSED	Astrophysical laboratory
EXP-A06	Rank $\leftrightarrow$ structural dimensionality	PROPOSED	Direct rank hypothesis

No experiment should yet be labeled **PREREGISTERED**, because several source-dependent details—including exact dataset versions, matching procedures, and some primary statistical metrics—still need to be frozen.

23. Required Next Registry

The next controlled registry should be the:

```
[  
  \boxed{  
    \textbf{Parameter Registry}  
  }  
]
```

It should freeze the numerical and computational choices that the Experiment Registry intentionally leaves open.

In particular, it must establish:

1. exact arithmetic dataset/version;
2. exact SDSS/astronomical dataset;
3. curve-selection rules;
4. rank computation method;
5. mapping normalization;
6. TDA implementation;
7. cosmic-web classifier;
8. SFR target definition;
9. train/validation/test partition;
10. null-generation algorithms;
11. primary statistical metrics;
12. effect-size thresholds;
13. multiple-testing procedure;
14. random seeds;
15. software versions.

Only after those fields are locked should the experiments transition:

```
[
\boxed{
\text{PROPOSED}
\rightarrow
\text{PREREGISTERED}.
}
]
```

24. Controlling Experimental Principle

The S.T.A.R. Experiment Registry adopts the following permanent rule:

A result cannot become evidence for a stronger claim merely because the analysis that produced it was successful.

Each result supports only the proposition actually tested.

Therefore:

```
[
\boxed{
\text{Experiment}
\rightarrow
\text{Claim}
}
]
```

is always explicit.

And:

```
[
\boxed{
\text{Claim}
\not\rightarrow
\text{Theory}
}
]
```

unless the intervening mathematical and empirical evidence has independently been established.

25. Registry Closing Statement

The Priority A program now has a controlled experimental architecture capable of distinguishing three very different outcomes:

[
\boxed{\text{interesting numerical correspondence}}
]

[
\boxed{\text{reproducible statistical relationship}}
]

[
\boxed{\text{independent predictive evidence}}.
]

Only the latter stages justify progressively stronger scientific interpretation.

The immediate goal is therefore not to demonstrate ACSC.

The immediate goal is to determine, under fixed and independently testable conditions:

[
\boxed{
\textbf{whether any arithmetic signal survives appropriate null models and independent astronomical validation.}
}
]

That result—positive, negative, or inconclusive—is the scientific output of the Priority A program.

Experiment Registry Initialization Record

Registry: S.T.A.R. Experiment Registry

Version: 0.1

Controlling Charter: Research Charter v0.2

Controlling Claim Registry: Claim/Evidence Registry v0.1

Priority: A

Status: PROPOSED

Experiments: EXP-A01 through EXP-A07

Next required controlled artifact:

```
[  
  \boxed{  
    \text{S.T.A.R. Parameter Registry v0.1}  
  }  
]
```

This registry will convert the remaining TBD – preregister before execution fields into fixed, auditable experimental parameters.